Upgrading Android A/B System from Android 13 to Android 14

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preface

Overview

This document describes the procedure to upgrade from Rockchip Android 13.0 A/B system to Android 14.0 A/B system via the over-the-air (OTA).

Product Version

Chip Name	Kernel Version
RK3326/RK3399/RK356x/RK3588/RK3562	5.10, 6.10

Intended Audience

This document (this guide) is mainly intended for

Technical Support Engineer

Software Development Engineer

Revision History

Version	Author	Date	Change Description
V1.0.0	Ji Dayao	2024-04-01	Initial version release

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1. Overview

This document describes the process of upgrading from the Rockchip Android 13.0 A/B system to the Android 14.0 A/B system via OTA.

2. Android 14.0 Patch

2.1 A/B Configuration

In the BoardConfig.mk of the corresponding project, enable the A/B configuration. The A/B configuration for Android 14 SDK must remain consistent with Android 13 A/B, meaning that if virtual A/B is not enabled in Android 13, the corresponding Android 14 should also not enable virtual A/B.

A/B Reference Configuration:

```
-BOARD_USES_AB_IMAGE := false
+BOARD_USES_AB_IMAGE := true
BOARD_ROCKCHIP_VIRTUAL_AB_ENABLE := false
```

Virtual A/B Reference Configuration:

```
-BOARD_USES_AB_IMAGE := false
-BOARD_ROCKCHIP_VIRTUAL_AB_ENABLE := false
+BOARD_USES_AB_IMAGE := true
+BOARD_ROCKCHIP_VIRTUAL_AB_ENABLE := true
```

2.2 Data Partition and Keymaster

1. First, follow the instructions in Section 3 "3. Upgrading from Android 13.0 to Android 14.0". If the system can boot into the Android main interface normally after the upgrade, it indicates that the data partition doesn't have problem, therefore, ignore the following point 2. If the system cannot boot into the Android main interface normally after the upgrade, proceed with the instructions described in point 2 below.
2. If the system fails to properly access the Android main interface after upgrade, retrieve the corresponding patch from the "userdata_patches" directory within the provided cloud drive link, then apply it to the corresponding repository in the Android 14 SDK development kit. For specific patch application procedures, please refer to the README.txt file located under the userdata_patches directory.

Link: https://pan.baidu.com/s/1HcH1QbUE4mAJplV5KoP-1w

Extraction code:0nsa

Note: This patch is only applicable for projects upgrading from old Android 13 devices to Android 14 devices. New devices developing Android 14 cannot use this patch.

3. Upgrading From Android 13.0 to Android 14.0

On the Rockchip Android 14.0 platform, follow the steps below to generate the corresponding OTA update package. Then place the update package on the Rockchip Android 13.0 device and perform a standard OTA update. Please read the “Notes” carefully before proceeding.

1. Select the corresponding lunch target for upgrading from 13.0 system to 14.0 system, such as lunch rk3588_t-userdebug or lunch rk3568_t-userdebug, etc.

Generally the corresponding lunch project directory can be directly copied from the Android 13 SDK.

2. Normally compile the system (compile uboot, kernel and after lunch execute the following commands for compilation):

```
make installclean && make -j16 && make dist -j16 && ./mkimage_ab.sh ota
```

3. Extract the upgrade package compiled in Step 2 and name it as update.zip.
4. Place the corresponding update.zip on the Android 13 device to perform the upgrade.
5. Notes:

- If the current device's Android 13.0 system was upgraded from an Android 12.0 system (via 《Rockchip_Developer_Guide_AB_System_OTA_from_Android12_to_Android13_CN》), then to further upgrade this Android 13.0 system to Android 14.0, the procedure follows 《Rockchip_Developer_Guide_AB_System_OTA_from_Android12_to_Android14_CN》.

4. Notes

1. During validation debugging for various Android platforms (e.g., Android13.0/Android14.0), please use the userdebug build variant and set BOARD_SELINUX_ENFORCING := false.
2. Ensure that the parameter file partition table is consistent before performing an OTA update. For example, when upgrading from Android 13.0 to Android 14.0, it is necessary to ensure that the parameter.txt file used in the compiled 14.0 firmware remains consistent with that of 13.0.
3. Before performing an OTA update, ensure that the file system and encryption method for the data partition in fstab (e.g., fstab.rk30board) must remain identical. For example, when upgrading from Android 13.0 to Android 14.0, it is necessary to ensure that the file system and encryption method used for the data partition in the compiled 14.0 firmware are identical to those of 13.0.
4. Before performing OTA updates, it is essential to ensure the compiled firmware itself works properly. For example, when upgrading from Android 13.0 to Android 14.0, the compiled 14.0 firmware must be verified through tools (such as RKDevTool) to ensure it can boot normally after being flashed onto the device, with all functions working properly.